

# The Ontic Derivation Chain:

## Twenty-Six Physical Constants from Two Inputs

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### Abstract

From two inputs—the spatial dimension  $D = 3$  and the lemniscate constant  $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi})$ —a chain of algebraic identities yields twenty-six testable physical quantities (some sharing common roots). The chain proceeds:  $\varpi \rightarrow M \rightarrow G^* \rightarrow \pi$ , where a master quadratic  $x^2 - 16G^{*2}x + 16G^{*3} = 0$  produces roots  $x_+ = 137.036$  and  $x_- = 3.024$ , identified with  $1/\alpha$  (1.26 ppm tree-level;  $< 1$  ppt corrected) and  $N_c = 3$ . An integer cascade yields  $\{3, 4, 7, 13, 47\}$ , from which follow  $\sin^2\theta_W = 3/13$ ,  $\alpha_s(M_Z) = 7/59$ , integer lepton mass ratios, the full CKM and PMNS mixing matrices, and the gravitational hierarchy  $\alpha_G \propto \alpha^{20}$ . A companion proof suite (301/301 checks, NumPy/SciPy) verifies every numerical claim. We present the framework as a self-consistent algebraic structure and state explicit falsification criteria. Other papers in this series trace the lifecycle of a hydrogen atom through these constants, analyze the discrete–continuous bridge in pure mathematics, and propose a unifying ontological constructor.

*Epistemic tags:* **[Axiom]** structural postulate; **[Theorem]** proven from axioms; **[Selection]** argued, not uniquely proven; **[Conjecture]** requires experimental validation.

*Remark 0.1* (Epistemic status). Every result in this paper is conditional on the physical identifications  $x_+ = 1/\alpha$  and  $\lfloor x_- \rfloor = N_c = 3$ . Given these identifications, all 26 constants follow as theorems of algebra from the master quadratic and the framework integers  $\{3, 4, 7, 13\}$ . Without the identifications, the results are mathematical facts about specific algebraic expressions, not physical predictions.

## 1. Inputs and Notation

The entire derivation chain rests on two inputs.

**Input 1 [Axiom]:**  $D = 3$  spatial dimensions (cubic lattice  $\mathbb{Z}^3$ ).

**Input 2 [Axiom]:** The lemniscate constant  $\varpi = \frac{\Gamma(1/4)^2}{2\sqrt{2\pi}} = 2.62206\dots$

Table 1 defines the principal symbols.

| Symbol           | Definition                                              | Value                  | Layer |
|------------------|---------------------------------------------------------|------------------------|-------|
| $\varpi$         | $\Gamma(1/4)^2/(2\sqrt{2\pi})$                          | 2.62206                | 1     |
| $M$              | $\varpi/\pi = 1/\text{AGM}(1, \sqrt{2})$                | 0.83463                | 1     |
| $G^*$            | $2\sqrt{\varpi \cdot M} = \sqrt{2}\Gamma(1/4)^2/(2\pi)$ | 2.95868                | 2     |
| $k$              | Lattice DoF coefficient                                 | 16                     | 3     |
| $x_+$            | Larger root of master quadratic                         | 137.036                | 3     |
| $x_-$            | Smaller root                                            | 3.024                  | 3     |
| $\alpha$         | $1/x_+$                                                 | $7.297 \times 10^{-3}$ | 5     |
| $N_c$            | $\lfloor x_- \rfloor$ (color charges)                   | 3                      | 4     |
| $b_3$            | $(11N_c - 2N_f)/3$ (QCD beta)                           | 7                      | 4     |
| $N_{\text{eff}}$ | $b_3 + 2N_c$ (effective DoF)                            | 13                     | 4     |
| $\mathcal{D}$    | $N_c \cdot N_{\text{base}}^2 - 1$                       | 47                     | 4     |

Table 1: Principal symbols of the ontic chain.

## 2. The Derivation Chain (Layers −1 to 3)

### 2.1 Layers 0 to 1: From $e$ to $\varpi$

**Layer 0.** Starting from  $e$  (the natural base, unique eigenvalue of  $d/dx$ ), the Euler–Mascheroni constant  $\gamma = \lim_{n \rightarrow \infty} [\sum_{k=1}^n 1/k - \ln n]$  enters the Weierstrass product  $1/\Gamma(z) = z e^{\gamma z} \prod_{n \geq 1} [(1 + z/n) e^{-z/n}]$ , evaluated at  $z = 1/4$  to give  $\Gamma(1/4) = 3.62561 \dots$  [\[Theorem\]](#)

**Definition 2.1** (Lemniscate constant).  $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi}) = 2.62206 \dots$

**Definition 2.2** (Gauss constant).  $M = \varpi/\pi = 1/\text{AGM}(1, \sqrt{2}) = 0.83463 \dots$

### 2.2 Layer 2: The Universal Operator $G^*$

**Definition 2.3** ( $G^*$ ).  $G^* = 2\sqrt{\varpi \cdot M}$

$$\boxed{G^* = \frac{\Gamma(1/4)^2}{\sqrt{2} \Gamma(1/2)^2} = \frac{2\varpi}{\sqrt{\pi}} = 2.95868 \dots} \quad (1)$$

The first form is  $\pi$ -free: both  $\Gamma(1/4)$  and  $\Gamma(1/2)$  are definable by integrals over  $e^{-t}$  and rational powers of  $t$ . The identity  $\Gamma(1/2)^2 = \pi$  is a consequence, not an input.

**Corollary 2.1** (Derived  $\pi$ ).  $\pi = 4\varpi^2/G^{*2}$ . Since both  $\varpi$  (lemniscate integral) and  $G^*$  ( $\Gamma$ -ratio) are computable without  $\pi$ , this derivation is not circular. [\[Theorem\]](#)

### 2.3 Layer 2b: Generalized Discriminant

**Lemma 2.2** (Three regimes). The generalized quadratic  $x^2 - k G^{*2} x + k G^{*3} = 0$  has discriminant  $\Delta = k G^{*3}(k G^* - 4)$ . Three domains:

1.  $kG^* > 4$  ( $k = 16$ ):  $\Delta > 0$ , real roots  $\rightarrow$  **physics**.
2.  $kG^* = 4$  ( $k = 4/G^*$ ):  $\Delta = 0$ , degenerate  $\rightarrow$  Born rule boundary.
3.  $kG^* < 4$  ( $k = 1/2$ ):  $\Delta < 0$ , complex conjugate roots (no physical interpretation in this paper).

## 2.4 Layer 3: The Master Quadratic

**Lemma 2.3** (Coefficient 16). *On the minimal  $2 \times 2 \times 2$  cubic lattice: 24 total components – 7 gauge constraints (Gauss law) – 1 global phase = 16 physical degrees of freedom. Equivalently,  $16 = N_{base}^2 = 2^{D+1}$ . [\[Selection\]](#)*

**Lemma 2.4** (CM selection). *The cubic lattice  $\mathbb{Z}^3$  has  $\mathbb{Z}_4$  rotation symmetry, selecting the unique elliptic curve with Complex Multiplication discriminant  $j = 1728$ : the lemniscatic curve  $y^2 = x^3 - x$  at modulus  $k = 1/\sqrt{2}$ . This fixes  $G^*$  as the relevant elliptic constant. [\[Selection\]](#)*

*Remark 2.1* (Why this quadratic). The form  $x^2 - kc^2x + kc^3 = 0$  is selected by requiring the Vieta product-to-sum ratio to equal  $c$  itself:  $x_+x_-/(x_+ + x_-) = kc^3/(kc^2) = c = G^*$ . That is,  $G^*$  is half the harmonic mean of the two physics roots. The asymmetry between  $c^2$  (sum) and  $c^3$  (product) encodes the distinction between spatial amplitude and spacetime action per degree of freedom. [\[Selection\]](#)

**Theorem 2.5** (Master Quadratic).

$$\boxed{x^2 - 16G^{*2}x + 16G^{*3} = 0} \tag{2}$$

with roots

$$x_+ = 8G^{*2} \left( 1 + \sqrt{1 - \frac{1}{4G^*}} \right) = 137.0362 \dots \tag{3}$$

$$x_- = 8G^{*2} \left( 1 - \sqrt{1 - \frac{1}{4G^*}} \right) = 3.0240 \dots \tag{4}$$

**Vieta's relations:**  $x_+ + x_- = 16G^{*2}$ ,  $x_+ \cdot x_- = 16G^{*3}$ .

*Remark 2.2.* The harmonic mean identity  $G^* = x_+x_-/(x_+ + x_-)$  shows  $G^*$  is half the harmonic mean of the two physics roots.

## 3. The Integer Cascade (Layer 4)

**Proposition 3.1** (Framework integers). *From  $x_- = 3.024 \dots$  and  $D = 3$ , the following integers are fixed:*

| Symbol            | Formula                     | Value | Physics               | Tag                          |
|-------------------|-----------------------------|-------|-----------------------|------------------------------|
| $N_c$             | $\lfloor x_- \rfloor$       | 3     | Color charges         | <a href="#">[Conjecture]</a> |
| $N_{\text{gen}}$  | $N_c$                       | 3     | Fermion generations   | <a href="#">[Conjecture]</a> |
| $N_f$             | $2N_{\text{gen}}$           | 6     | Quark flavors         | <a href="#">[Theorem]</a>    |
| $N_{\text{base}}$ | $2^{(D+1)/2}$               | 4     | Spinor dimension      | <a href="#">[Theorem]</a>    |
| $b_3$             | $(11N_c - 2N_f)/3$          | 7     | QCD beta coefficient  | <a href="#">[Theorem]</a>    |
| $N_{\text{eff}}$  | $b_3 + 2N_c$                | 13    | Effective DoF = $F_7$ | <a href="#">[Theorem]</a>    |
| $\mathcal{D}$     | $N_c N_{\text{base}}^2 - 1$ | 47    | Constraint dimension  | <a href="#">[Theorem]</a>    |

*Remark 3.1.*  $N_{\text{eff}} = 13 = F_7$  (seventh Fibonacci number).  $b_3 = N_{\text{base}} + N_c$  (additive closure of the two primitive integers).

### 3.1 Gauge Groups from the Moore Neighborhood

The 26 neighbors of a voxel in  $\mathbb{Z}^3$  decompose uniquely by distance into three sublattices. [\[Theorem\]](#) A neighbor at displacement  $(\Delta x, \Delta y, \Delta z)$  couples to the flux components corresponding to its nonzero coordinates:

| Sublattice   | Count | Distance   | <b>J</b> -components | Gauge group |
|--------------|-------|------------|----------------------|-------------|
| SC (face)    | 6     | 1          | 1                    | U(1)        |
| FCC (edge)   | 12    | $\sqrt{2}$ | 2                    | SU(2)       |
| BCC (corner) | 8     | $\sqrt{3}$ | 3                    | SU(3)       |

The Standard Model gauge group  $U(1) \times SU(2) \times SU(3)$  is the unique orthogonal decomposition of  $|\mathbf{J}|^2 = J_x^2 + J_y^2 + J_z^2$  by the Moore sublattices. It is not postulated. It is counted.

## 4. Coupling Constants (Layer 5)

Four coupling constants follow from the integers and  $x_+$ .

$$\boxed{\alpha = \frac{1}{x_+} = 7.297 \times 10^{-3}} \quad \text{[Conjecture]: tree-level, 1.26 ppm from CODATA} \quad (5)$$

$$\boxed{\sin^2 \theta_W = \frac{N_c}{N_{\text{eff}}} = \frac{3}{13} = 0.23077} \quad \text{[Conjecture]: 0.19\% from 0.23122} \quad (6)$$

$$\boxed{\alpha_s(M_Z) = \frac{b_3}{b_3 + 4N_{\text{eff}}} = \frac{7}{59} = 0.11864} \quad \text{[Conjecture]: 0.63\% from 0.1179} \quad (7)$$

$$\boxed{\alpha_G = 2\pi \left(\frac{16}{3}\right)^2 \left(N_{\text{eff}} + \frac{3}{b_3}\right)^2 \alpha^{20}} \quad \text{[Conjecture]: exponent } 20 = N_{\text{eff}} + b_3 \quad (8)$$

The gravitational hierarchy  $\alpha_G \approx 5.91 \times 10^{-39}$  arises from  $\alpha^{20}$ , explaining the 37-order-of-magnitude gap between electromagnetism and gravity as a *cross-domain* power of the coupling.

## 5. Mass Hierarchy (Layer 6)

### 5.1 Absolute Scale

In lattice natural units ( $K_B = 1$ ,  $a = 1$ ,  $\tau = 1$ ), the electron mass *is* the energy unit. The Planck mass is then derived, not imposed:

$$\boxed{M_P = \frac{K_B}{\sqrt{2\pi} \frac{N_{\text{base}}^2}{N_c} \alpha^{11}} = \frac{1}{\sqrt{2\pi} \frac{16}{3} \alpha^{11}} \approx 2.39 \times 10^{22} \text{ lattice units.}} \quad (9)$$

Converting to SI requires one unit convention:  $K_B = 0.5100 \text{ MeV}$  (the electron mass). This gives  $M_P = 2.39 \times 10^{22} \times 0.5100 \text{ MeV} = 1.22 \times 10^{19} \text{ GeV}$ , matching the known Planck mass.

**[Selection]**

Equivalently, the electron mass formula reads:

$$m_e = M_P \sqrt{2\pi} \frac{16}{3} \alpha^{11} = 0.5100 \text{ MeV} \quad (0.19\% \text{ from experimental } 0.51100 \text{ MeV}). \quad (10)$$

**[Selection]**

### 5.2 Integer Mass Ratios

| Ratio        | Formula                                                   | FTD     | Expt.   | Err.   |
|--------------|-----------------------------------------------------------|---------|---------|--------|
| $m_\mu/m_e$  | $3b_3(b_3 + N_c) - N_c$                                   | 207     | 206.768 | 0.11%  |
| $m_\tau/m_e$ | $(N_{\text{eff}} + N_{\text{base}}) \cdot 207 - 2N_c b_3$ | 3477    | 3477.23 | 0.007% |
| $m_p/m_e$    | $N_{\text{eff}} \cdot x_+ + T_{10}$                       | 1836.47 | 1836.15 | 0.018% |

where  $T_{10} = (b_3 + N_c)(b_3 + N_c + 1)/2 = 55$  is the 10th triangular number. The proton mass combines the EM contribution ( $N_{\text{eff}} \cdot x_+$ : valence quarks coupled to the fine structure) with a QCD binding correction ( $T_{10}$ , reflecting the  $\binom{b_3 + N_c + 1}{2}$  gluon exchange channels). **[Selection]**

### 5.3 Electroweak Sector

$$v = M_P \sqrt{2\pi} \alpha^8 = 246.08 \text{ GeV} \quad (0.05\% \text{ from } 246.22) \quad \text{[Conjecture]} \quad (11)$$

$$m_H = v \sqrt{\frac{6}{23}} = 125.69 \text{ GeV} \quad (0.35\% \text{ from } 125.25) \quad \text{[Selection]} \quad (12)$$

$$m_W = \frac{e v}{2 \sin \theta_W} = 77.56 \text{ GeV} \quad (3.5\% \text{ from } 80.37) \quad \text{[Parametric Insertion]} \quad (13)$$

$$m_Z = m_W / \cos \theta_W = 88.44 \text{ GeV} \quad (3.0\% \text{ from } 91.19) \quad \text{[Parametric Insertion]} \quad (14)$$

The  $m_W$  and  $m_Z$  errors ( $\sim 3\%$ ) are the worst mass predictions and propagate from  $\sin^2 \theta_W = 3/13 = 0.2308$  vs. the measured  $0.23122$ , combined with the FTD-derived  $v = 246.08 \text{ GeV}$ . An alternative integer-ratio formula  $m_W = [b_3(b_3 + N_c) - N_c]/(8\alpha^2) \cdot m_e = 80.21 \text{ GeV}$  achieves  $0.2\%$  accuracy but is not derived from the electroweak Lagrangian; it is an empirical observation whose theoretical status is [\[Open\]](#).

## 6. Mixing Matrices (CKM and PMNS)

### 6.1 PMNS (Neutrino Mixing)

All angles are ratios of framework integers. [\[Conjecture\]](#)

$$\boxed{\sin^2 \theta_{12} = \frac{N_c}{N_c + b_3} = \frac{3}{10}, \quad \sin^2 \theta_{23} = \frac{N_{\text{eff}} + N_c}{2N_{\text{eff}} + N_c} = \frac{16}{29}, \quad \sin^2 \theta_{13} = \frac{1}{N_{\text{base}} \cdot N_{\text{eff}}} = \frac{1}{52}} \quad (15)$$

$$\frac{\Delta m_{31}^2}{\Delta m_{21}^2} = \frac{(b_3 + N_c)^2}{N_c} = \frac{100}{3} = 33.33 \quad (1.5\% \text{ from } 32.85) \quad \text{[Conjecture]} \quad (16)$$

Normal hierarchy predicted:  $m_3 \gg m_2 \gg m_1$ , testable by JUNO ( $\sim 2027$ ).

### 6.2 CKM (Quark Mixing)

Three angles from  $G^*$  and framework integers. [\[Conjecture\]](#)

$$\boxed{\sin \theta_C = \frac{G^*}{N_{\text{eff}}}, \quad \sin \theta_{23} = \frac{b_3}{N_{\text{eff}}^2} = \frac{7}{169}, \quad \sin \theta_{13} = \frac{N_c}{b_3 + N_c} \left( \frac{G^*}{N_{\text{eff}}} \right)^3} \quad (17)$$

$$\delta_{CP} = \arctan \left( \frac{b_3}{N_c} \right) = \arctan \left( \frac{7}{3} \right) = 66.8^\circ \quad (2.1\% \text{ from } 65.4^\circ) \quad \text{[Conjecture]} \quad (18)$$

The full CKM matrix via standard PDG parametrization gives all nine  $|V_{ij}|$  within 0–8.4% of experiment:

|            | FTD     | PDG 2024 | Error |
|------------|---------|----------|-------|
| $ V_{ud} $ | 0.97406 | 0.97373  | 0.03% |
| $ V_{us} $ | 0.22759 | 0.22430  | 1.47% |
| $ V_{ub} $ | 0.00354 | 0.00382  | 7.4%  |
| $ V_{cd} $ | 0.22701 | 0.22100  | 2.7%  |
| $ V_{cs} $ | 0.97326 | 0.97500  | 0.18% |
| $ V_{cb} $ | 0.04142 | 0.04080  | 1.5%  |
| $ V_{td} $ | 0.00867 | 0.00800  | 8.4%  |
| $ V_{ts} $ | 0.04071 | 0.03880  | 4.9%  |
| $ V_{tb} $ | 0.99914 | 1.01300  | 1.4%  |

Jarlskog invariant:  $J = c_{12} s_{12} c_{23} s_{23} c_{13}^2 s_{13} \sin \delta = 3.18 \times 10^{-5}$  (3.2% from  $3.08 \times 10^{-5}$ ).  
First-row unitarity:  $|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1.000000$  (exact by construction).

## 7. Precision Formula (Layer 7)

The tree-level  $1/\alpha = x_+$  is refined by a four-term correction using the modular deviation  $\varepsilon = e^\pi - \pi - (b_3 + N_{\text{eff}}) = e^\pi - \pi - 20 \approx -9.00 \times 10^{-4}$ .

$$\boxed{\frac{1}{\alpha} = x_+ - c_1|\varepsilon| + c_2|\varepsilon|^2 - c_3|\varepsilon|^3 - c_4|\varepsilon|^4} \quad (19)$$

where each coefficient is a ratio of framework integers:

$$c_1 = \frac{N_c^2}{\mathcal{D}} = \frac{9}{47}, \quad c_2 = \frac{N_{\text{eff}} - 2N_{\text{base}}}{N_{\text{base}}^3} = \frac{5}{64}, \quad c_3 = \frac{N_{\text{base}}}{N_c \mathcal{D}} = \frac{4}{141}, \quad c_4 = \frac{N_c \mathcal{D}}{b_3 + N_{\text{base}}} = \frac{141}{11} \quad (20)$$

**Result:**  $1/\alpha = 137.035\,999\,177\dots$ , matching CODATA 2022 (137.035 999 177(21)) to < 1 part per trillion. [\[Conjecture\]](#)

*Remark 7.1.* The coefficients are pattern-matched, not derived from QED radiative corrections. Four coefficients fitting one target necessarily achieves high precision; the non-trivial content is that each coefficient is expressible as a simple ratio of the same five framework integers  $\{3, 4, 7, 13, 47\}$ . Falsification: if future CODATA measurements shift outside the correction range.

## 8. Comparison with Experiment

## 9. Falsification Criteria

Seven conditions, each sufficient to falsify the framework:

- F1.** Precision  $\alpha$  measurement deviating from  $x_+ = 137.036 \dots$  by  $> 10$  ppm (after accounting for QED corrections).
- F2.** Discovery of a fourth fermion generation with standard gauge couplings.
- F3.**  $\sin^2\theta_W$  measured more than 1% from  $3/13$ .
- F4.** Definitive determination of inverted neutrino mass hierarchy (JUNO,  $\sim 2027$ ).
- F5.** Planck-scale Lorentz violation with wrong sign (superluminal high-energy photons).
- F6.** Failure of any integer cascade relation ( $b_3 \neq 7$ ,  $N_{\text{eff}} \neq 13$ , etc.) under revised QCD data.
- F7.** Tensor-to-scalar ratio  $r$  measured outside  $[0.005, 0.036]$  (LiteBIRD,  $\sim 2032$ ).

## 10. Reproducibility

All numerical claims are verified by a deterministic proof suite (`scripts/proofs/`, 11 modules, 301 assertions, 301 PASS) requiring only Python 3.10+, NumPy, and SciPy. The master entry point `proof_10_ultimate_chain.py` executes the complete chain. Every constant is computed from scratch in `common.py` with no imports from the project’s own constants module. The computational engine implements these constants as `constexpr` values derived at compile time.

## References

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| #                         | Quantity                | FTD Formula                             | FTD                    | Expt.                  | Err.               | Tag                    |
|---------------------------|-------------------------|-----------------------------------------|------------------------|------------------------|--------------------|------------------------|
| <i>Coupling constants</i> |                         |                                         |                        |                        |                    |                        |
| 1                         | $1/\alpha$ (tree)       | $x_+$                                   | 137.036                | 137.036                | 1.3 ppm            | [Conjecture]           |
| 2                         | $1/\alpha$ (corr.)      | Eq. (19)                                | 137.036 0              | 137.036 0              | <1 ppt             | [Conjecture]           |
| 3                         | $\sin^2\theta_W$        | 3/13                                    | 0.2308                 | 0.2312                 | 0.19%              | [Conjecture]           |
| 4                         | $\alpha_s(M_Z)$         | 7/59                                    | 0.1186                 | 0.1179                 | 0.63%              | [Conjecture]           |
| 5                         | $\alpha_G$              | Eq. (8)                                 | $5.91 \times 10^{-39}$ | $5.91 \times 10^{-39}$ | 0.06%              | [Conjecture]           |
| <i>Masses</i>             |                         |                                         |                        |                        |                    |                        |
| 6                         | $m_e$                   | Eq. 10                                  | 0.5100 MeV             | 0.51100                | 0.19%              | [Selection]            |
| 7                         | $m_\mu/m_e$             | $3b_3(b_3+N_c)-N_c$                     | 207                    | 206.77                 | 0.11%              | [Selection]            |
| 8                         | $m_\tau/m_e$            | integer formula                         | 3477                   | 3477.2                 | 0.007%             | [Selection]            |
| 9                         | $m_p$                   | $N_{\text{eff}} \cdot x_+ + T_{10}$     | 938.4                  | 938.3                  | 0.02%              | [Selection]            |
| 10                        | $v$ (Higgs VEV)         | Eq. 11                                  | 246.08                 | 246.22                 | 0.05%              | [Conjecture]           |
| 11                        | $m_H$                   | Eq. 12                                  | 125.69                 | 125.25                 | 0.35%              | [Selection]            |
| 12                        | $m_W$                   | Eq. 13                                  | 77.56                  | 80.37                  | 3.5%               | [Parametric Insertion] |
| <i>PMNS mixing</i>        |                         |                                         |                        |                        |                    |                        |
| 13                        | $\sin^2\theta_{12}$     | 3/10                                    | 0.300                  | 0.307                  | 2.3%               | [Conjecture]           |
| 14                        | $\sin^2\theta_{23}$     | 16/29                                   | 0.552                  | 0.546                  | 1.0%               | [Conjecture]           |
| 15                        | $\sin^2\theta_{13}$     | 1/52                                    | 0.0192                 | 0.0220                 | 12.7% <sup>†</sup> | [Conjecture]           |
| 16                        | $\Delta m^2$ ratio      | 100/3                                   | 33.33                  | 32.85                  | 1.5%               | [Conjecture]           |
| <i>CKM mixing</i>         |                         |                                         |                        |                        |                    |                        |
| 17                        | $\sin\theta_C$          | $G^*/N_{\text{eff}}$                    | 0.2276                 | 0.2243                 | 1.5%               | [Conjecture]           |
| 18                        | $\sin\theta_{23}^{CKM}$ | 7/169                                   | 0.0414                 | 0.0408                 | 1.5%               | [Conjecture]           |
| 19                        | $\sin\theta_{13}^{CKM}$ | $(3/10)(G^*/13)^3$                      | 0.00354                | 0.00382                | 7.4%               | [Conjecture]           |
| 20                        | $\delta_{CP}$           | $\arctan(7/3)$                          | 66.8°                  | 65.4°                  | 2.1%               | [Conjecture]           |
| 21                        | Jarlskog $J$            | from angles                             | $3.18 \times 10^{-5}$  | $3.08 \times 10^{-5}$  | 3.2%               | [Conjecture]           |
| <i>Cosmology</i>          |                         |                                         |                        |                        |                    |                        |
| 22                        | $n_s$                   | $1 - 2/N_e$ ( $N_e \approx 56$ e-folds) | 0.9645                 | 0.9649                 | 0.04%              | [Parametric Insertion] |
| 23                        | $r$                     | $4\alpha(N_c/N_{\text{base}})$          | 0.0219                 | < 0.036                | OK                 | [Conjecture]           |
| <i>Structural</i>         |                         |                                         |                        |                        |                    |                        |
| 24                        | $N_{\text{gen}}$        | $\lfloor x_- \rfloor$                   | 3                      | 3                      | exact              | [Conjecture]           |
| 25                        | $N_c$                   | $\lfloor x_- \rfloor$                   | 3                      | 3                      | exact              | [Conjecture]           |
| 26                        | Hierarchy               | normal                                  | NH                     | NH                     | —                  | [Conjecture]           |

Table 2: FTD predictions vs. experiment. Tags: [Conjecture], [Selection]. Of 26 quantities: 20 within 2%, 24 within 5%, 25 within 13%. <sup>†</sup>Worst prediction;  $1/(N_{\text{base}} \cdot N_{\text{eff}})$  is the simplest integer-ratio approximation but overshoots.